

88 Speedmart System

Comprehensive Presentation Guide & Marking Rubric Defense BMCS3183 Advanced Database Management

1. Executive Overview & Member Module Allocations

This guide provides step-by-step presentation scripts, technical rubrics, and Q&A; defense points for each of the **5 team members**. Tasks 4 through 9 represent individually assessed components accounting for **65% of the final coursework grade**.

Member	Assigned Module & Domain	Source Script Path
Member 1	Module 1: Member & Loyalty Management	modules/01_member_loyalty/member_loyalty.sql
Member 2	Module 2: Inventory & Purchasing Management	modules/02_inventory_purchasing/inventory_purchasing.sql
Member 3	Module 3: Order, Payment & Fulfillment	modules/03_order_fulfillment/order_fulfillment.sql
Member 4	Module 4: Promotion & Marketing Management	modules/04_promotions/promotions.sql
Member 5	Module 5: Refund & Return (Selected Extra Module)	modules/05_refund_return/refund_return.sql

2. Individual Marking Rubric Checklist (Tasks 4 to 9)

Task & Weight	Rubric Requirement	Demonstration Technique for Presentation
Task 4: Queries (10%)	2 Multi-Table Queries (Strategic + Tactical levels). Single table queries forbidden.	Execute query in SQL*Plus/Developer; explain business metrics (churn risk, spend, capacity).
Task 5: Stored Procedures (10%)	2 Stored Procedures with parameters, validation, DML & error handling.	Run with valid args; then pass invalid args to showcase custom exceptions and rollback.
Task 6: Triggers (10%)	2 Conditional Triggers (with explicit WHEN condition).	Attempt illegal DML to prove trigger fires and aborts unauthorized transaction.
Task 7: Reports (20%)	2 Reports using Nested Cursors (Max rubric tier: 8 marks+).	Execute report procedure; show ASCII table formatting, line items & running totals.
Task 8: Extra Efforts (10%)	1 Sequence, 2 Views, 2 Indexes, 2 Formatting, 2 Exception types (PRAGMA).	Show DDL and explain indexing strategy & PRAGMA exception binding.
Task 9: Presentation (15%)	Live execution, fluent explanation, and Q&A; defense.	Follow the prepared speaking points and answer technical questions confidently.

Member 1: Module 1 — Member & Loyalty Management

A. Rubric Deliverables & Marks Pointers

- **Task 8 (Extra Efforts):** seq_member_id (starts 1000); Indexes: idx_member_status_type, idx_redemption_member_order; Views: v_member_loyalty_summary (Strategic), v_voucher_utilization (Tactical); Exceptions: Custom e_insufficient_points, e_invalid_email + PRAGMA EXCEPTION_INIT(e_foreign_key_violation, -2291).
- **Task 4 (Queries):** Query 1 (Strategic: Member churn risk matrix by inactivity months); Query 2 (Tactical: Voucher claim conversion rates and point burn rate).
- **Task 5 (Procedures):** sp_register_or_renew_member (validates email, calculates 12-month VIP expiry); sp_process_point_redemption (validates voucher rules, point balance deduction).
- **Task 6 (Triggers):** trg_guard_point_redemption (blocks inactive members/insufficient balance); trg_enforce_vip_expiration (auto-sets 1-year expiry on VIP upgrade).
- **Task 7 (Nested Cursor Reports):** rpt_member_annual_statement(1) (Parent: Member → Child 1: Orders → Child 2: Redemptions); rpt_voucher_performance_summary (Parent: Vouchers → Child: Member claims).

B. Live Demonstration Script

```
SET SERVEROUTPUT ON SIZE UNLIMITED;
-- 1. Register VIP Member (Computes 1-Year Expiry)
VARIABLE v_new_mem NUMBER;
EXEC sp_register_or_renew_member('Alice Walker', 'alice.w@test.com', 'pwd123', '012-1122334', 'VIP', '14
Jalan Ampang, KL', :v_new_mem);
PRINT v_new_mem;

-- 2. Process Point Redemption for Order
VARIABLE v_red_id NUMBER;
EXEC sp_process_point_redemption(1, 1, 1, :v_red_id);

-- 3. Execute Nested Cursor Reports
EXEC rpt_member_annual_statement(1);
EXEC rpt_voucher_performance_summary;
```

C. Expected Lecturer Q&A; Defense

Q: Why use PRAGMA EXCEPTION_INIT in your procedure?

A: It binds Oracle's internal foreign key violation code (-2291) to a named exception e_foreign_key_violation, allowing clean transactional rollbacks and user-friendly error messages without crashing.

Q: How do you prevent nested cursor memory leaks?

A: We use explicit OPEN/FETCH/CLOSE for the parent header and PL/SQL FOR rec IN c_orders(member_id) loops for children, which automatically handle cursor lifecycle.

Member 2: Module 2 — Inventory & Purchasing Management

A. Rubric Deliverables & Marks Pointers

- **Task 8 (Extra Efforts):** seq_po_id (500), seq_stock_log_id (2000); Indexes: idx_stock_branch_reorder, idx_batch_item_expiry; Views: v_branch_inventory_health (Strategic), v_supplier_procurement_summary (Tactical); Exceptions: Custom e_po_not_approved, e_employee_unauthorized + PRAGMA EXCEPTION_INIT(e_check_constraint_violated, -2290).
- **Task 4 (Queries):** Query 1 (Strategic: Branch replenishment deficiency & reorder cost); Query 2 (Tactical: Batches expiring <90 days & value at risk).
- **Task 5 (Procedures):** sp_receive_purchase_order (checks staff store auth, MERGE stock upsert, batch creation, StockLog audit); sp_adjust_damaged_stock (decrements quantity and logs write-off).
- **Task 6 (Triggers):** trg_guard_po_item_integrity (enforces positive cost/qty); trg_guard_maximum_stock_capacity (blocks updates exceeding MaximumStock limit).
- **Task 7 (Nested Cursor Reports):** rpt_branch_inventory_audit(1) (Parent: Branch → Child 1: Stock levels & alerts → Child 2: Write-offs); rpt_supplier_procurement_audit(1) (Parent: Supplier → Child 1: POs → Child 2: Line items).

B. Live Demonstration Script

```
SET SERVEROUTPUT ON SIZE UNLIMITED;
-- 1. Receive PO #1 by Branch 1 Manager (Staff #7)
EXEC sp_receive_purchase_order(1, 7, 1);

-- 2. Adjust Damaged Inventory
EXEC sp_adjust_damaged_stock(1, 1, 7, 2, 'Damaged', 'Dropped carton during shelf placement');

-- 3. Execute Nested Cursor Reports
EXEC rpt_branch_inventory_audit(1);
EXEC rpt_supplier_procurement_audit(1);
```

C. Expected Lecturer Q&A; Defense

Q: Why use MERGE INTO when receiving purchase orders?

A: MERGE INTO atomically updates quantities for existing stock records and inserts new rows for first-time stocked items in a single statement, avoiding race conditions.

Q: How do you enforce branch staff authorization?

A: In sp_receive_purchase_order, we check v_emp_branch <> p_branch_id and raise e_employee_unauthorized if a staff member attempts to receive goods for a different store.

Member 3: Module 3 — Order, Payment & Fulfillment

A. Rubric Deliverables & Marks Pointers

- **Task 8 (Extra Efforts):** seq_order_id (600), seq_payment_id (800); Indexes: idx_order_date_status, idx_payment_method_stat; Views: v_order_fulfillment_summary (Strategic), v_courier_delivery_efficiency (Tactical); Exceptions: Custom e_order_not_pending + PRAGMA EXCEPTION_INIT(e_duplicate_transaction, -1) for transaction numbers.
- **Task 4 (Queries):** Query 1 (Strategic: Omnichannel revenue split between Delivery vs In-Store Pickup); Query 2 (Tactical: Courier partner SLA success rate and freight volume).
- **Task 5 (Procedures):** sp_create_pickup_order (generates 6-digit PickupCode, creates order/pickup atomically); sp_settle_order_payment (settles invoice, marks order Completed, awards 1 point per RM 1.00 spent).
- **Task 6 (Triggers):** trg_guard_exclusive_delivery (blocks Delivery insertion if Pickup exists); trg_guard_paid_payment_state (prevents reversing Paid to Pending).
- **Task 7 (Nested Cursor Reports):** rpt_order_tax_invoice(1) (Parent: Order & Customer & Pickup/Delivery → Child: Line items & savings); rpt_branch_daily_manifest(1) (Parent: Branch → Child 1: Pickups with codes → Child 2: Deliveries).

B. Live Demonstration Script

```
SET SERVEROUTPUT ON SIZE UNLIMITED;
-- 1. Create Pickup Order (Generates 6-Digit Claim Code)
VARIABLE v_ord NUMBER;
VARIABLE v_code VARCHAR2(10);
EXEC sp_create_pickup_order(1, 1, :v_ord, :v_code);
PRINT v_ord;
PRINT v_code;

-- 2. Settle Payment & Award Points
EXEC sp_settle_order_payment(:v_ord, 'Online Banking', 145.00, 'TXN-MBB-998811');

-- 3. Execute Nested Cursor Reports
EXEC rpt_order_tax_invoice(:v_ord);
EXEC rpt_branch_daily_manifest(1);
```

C. Expected Lecturer Q&A; Defense

Q: How is the business rule of mutually exclusive fulfillment enforced?

A: By trg_guard_exclusive_delivery, which verifies no corresponding record exists in Pickup before allowing insertion into Delivery, raising error -20210.

Q: How is loyalty point accrual handled during payment?

A: In sp_settle_order_payment, upon payment commit, we calculate TRUNC(AmountPaid) and update Member.MemberPoint in the same transactional unit.

Member 4: Module 4 — Promotion & Marketing Management

A. Rubric Deliverables & Marks Pointers

- **Task 8 (Extra Efforts):** seq_promo_id (300); Indexes: idx_promo_date_range, idx_promo_item_lookup; Views: v_active_promotion_catalog (Strategic), v_promotion_sales_performance (Tactical); Exceptions: Custom e_invalid_dates, e_negative_discount + PRAGMA EXCEPTION_INIT(e_numeric_overflow, -1438).
- **Task 4 (Queries):** Query 1 (Strategic: Campaign revenue uplift & absorbed discount cost); Query 2 (Tactical: Deepest discount deals & percentage markdowns).
- **Task 5 (Procedures):** sp_create_promotional_campaign (validates dates, positive discount, checks base price); sp_apply_order_promo_discount (batch updates pending order lines with active promo discounts).
- **Task 6 (Triggers):** trg_guard_promotion_date_range (prohibits EndDate <= StartDate); trg_guard_promo_item_discount (prohibits discount >= Item Price).
- **Task 7 (Nested Cursor Reports):** rpt_active_promotions_catalog (Parent: Campaigns → Child: Enrolled items with original vs promo prices); rpt_promotion_sales_audit(1) (Parent: Campaign → Child: Order lines & customer savings).

B. Live Demonstration Script

```
SET SERVEROUTPUT ON SIZE UNLIMITED;
-- 1. Create Promotion Campaign with Item #2 enrolled
VARIABLE v_p_id NUMBER;
EXEC sp_create_promotional_campaign(2.00, SYSDATE, SYSDATE + 14, 2, :v_p_id);
PRINT v_p_id;

-- 2. Apply Discounts to Pending Order
EXEC sp_apply_order_promo_discount(1);

-- 3. Execute Nested Cursor Reports
EXEC rpt_active_promotions_catalog;
EXEC rpt_promotion_sales_audit(1);
```

C. Expected Lecturer Q&A; Defense

Q: How do you prevent promotional discounts from generating negative prices?

A: Both the stored procedure and trigger trg_guard_promo_item_discount compare DiscountAmount against Item.Price, rejecting any promotion where discount is greater than or equal to price.

Q: Why use view v_active_promotion_catalog?

A: It abstracts temporal window filtering (SYSDATE BETWEEN StartDate AND EndDate) and pre-computes promotional prices and discount percentages for fast customer queries.

Member 5: Module 5 — Refund & Return Management (Extra Module)

A. Rubric Deliverables & Marks Pointers

- **Task 8 (Extra Efforts):** seq_refund_id (1500); Indexes: idx_refund_order_stat, idx_returnitem_lookup; Views: v_refund_claim_summary (Strategic), v_defective_item_loss (Tactical); Exceptions: Custom e_exceeded_7_days, e_invalid_condition + PRAGMA EXCEPTION_INIT(e_check_constraint, -2290).
- **Task 4 (Queries):** Query 1 (Strategic: Quality defect loss ranking to trigger supplier audits); Query 2 (Tactical: Branch refund claim frequencies and payout rates).
- **Task 5 (Procedures):** sp_submit_refund_claim (enforces 7-day window [Rule 31] & item condition [Rule 33]); sp_adjudicate_refund (approves claim, updates Payment to Refunded [Rule 32] and OrderDetail line to Returned).
- **Task 6 (Triggers):** trg_guard_return_item_condition (blocks non-flawed returns); trg_guard_refund_amount_limit (prohibits refund exceeding order payment).
- **Task 7 (Nested Cursor Reports):** rpt_refund_claim_dossier(1) (Parent: Claim & Customer & Photo → Child: Returned items & remarks); rpt_branch_quality_audit(1) (Parent: Branch → Child 1: Claims → Child 2: Items).

B. Live Demonstration Script

```
SET SERVEROUTPUT ON SIZE UNLIMITED;
-- 1. Lodge Return Claim & Adjudicate
VARIABLE v_ref_id NUMBER;
EXEC sp_submit_refund_claim(2, 'Milk bottle seal broken and leaked', 'Drop-off at Branch',
'http://img.freshmart/p1.jpg', 1, 1, 'Damaged', :v_ref_id);
PRINT v_ref_id;

EXEC sp_adjudicate_refund(:v_ref_id, 'Approved', 'Verified at store counter; refund approved.');
```

```
-- 2. Execute Nested Cursor Reports
EXEC rpt_refund_claim_dossier(:v_ref_id);
EXEC rpt_branch_quality_audit(1);
```

C. Expected Lecturer Q&A; Defense

Q: How is the 7-day refund policy enforced?

A: In sp_submit_refund_claim, we calculate (SYSDATE - OrderDate) > 7. If exceeded, custom exception e_exceeded_7_days is raised with error -20404.

3. Presentation Day Practical Checklist

1. Open Oracle SQL Developer / SQL*Plus and execute @Dataset.txt to refresh tables and data.
2. Execute @run_all_modules.sql to compile all 5 modules.
3. Run SET SERVEROUTPUT ON SIZE UNLIMITED; before calling any report procedure.
4. Demonstrate both success and intentional error cases to showcase custom exception handling.